

We thank you for your time spent taking this survey.
Your response has been recorded.

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responses

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Name:

Key

Email Address:

Key

Use the "harass" data set to examine if the effect of harassment at time 1 (harass) on depression at time 2 (dep2) is mediated through failure at time 2 (fail2). Control for failure at time 1 (fail1) and depression at time 1 (dep1) in your analysis. Use PROCESS or regression in SPSS, SAS, or R to estimate and test this question.

What variable is the independent variable (X)?

- ☒ **harassment at time 1 (harass)**
- ☐ depression at time 2 (dep2)
- ☐ failure at time 2 (fail2)

What variable is the outcome variable (Y)?

- ☐ harassment at time 1 (harass)
- ☒ **depression at time 2 (dep2)**
- ☐ failure at time 2 (fail2)

What variable is the mediator variable (M)?

- ☐ harassment at time 1 (harass)

☐ depression at time 2 (dep2)

☒ failure at time 2 (fail2)

Paste the code that you would use to estimate this model.

```
process(data = harass, y = "dep2", x = "harass", m = "fail2", cov = c("fail1", "dep1"), model = 4, total = 1)
```

What is the estimate of the a-path, or the effect of the independent variable (X) on the mediator (M)?

☐ .1413

☐ -.0121

☒ .0857

☐ .2644

Provide a one sentence interpretation of what the a-path means in this context (make sure to include controls in the interpretation)

```
Among students two started out the same in depression and perceived academic failure, a student who experienced 1 unit more of harassment is expected to have .0857 units higher perceived academic failure 1 year later.
```

What is the estimate of the b-path, or the effect of the mediator (M) on the outcome (Y), controlling for (X)?

☒ .2239

☐ .0761

☐ .2644

☐ -.0121

Provide a one sentence interpretation of what the b-path means in this context (make sure to include controls in the interpretation)

```
For individuals who started out the same on depression and perceived academic failure and have the same level of harassment, if one student experiences 1 unit more of perceived academic failure at follow-up that student is expected to be .2239 units higher on depression at follow-up as well.
```

What is the estimate of the direct effect (c'-path), or the effect of the independent variable (X) on the outcome (Y), controlling for M?

- ☒ **-.0121**
- ☐ .0192
- ☐ .0857
- ☐ .4675

Provide a one sentence interpretation of what the direct effect means in this context (make sure to include controls in the interpretation)

Among those two started off the same on depression and perceived academic failure and are the same on perceived academic failure at follow-up, if one student is 1 unit higher on harassment that student is expected to be .0121 units lower on depression at follow-up.

What is the estimate of the total effect (c-path), or the effect of the independent variable (X) on the outcome (Y)?

- ☒ **.0071**
- ☐ .1352
- ☐ .4183
- ☐ .9676

Provide a one sentence interpretation of what the total effect means in this context (make sure to include controls in the interpretation)

For two individuals who started out the same on depression and perceived academic failure, if one individual is one unit higher on harassment, then they are expected to be .0071 units higher on depression at follow-up.

What is the estimate of the indirect effect, or the effect of the independent variable (X) on the the outcome (Y) through (M)?

- ☒ **.0192**
- ☐ -.0121
- ☐ .0071
- ☐ .2239

Provide a one sentence interpretation of what the indirect effect means in this context (make sure to include controls in the interpretation)

For two individuals who started the same on depression and perceived academic failure but differ by 1 unit on harassment, they are expected to differ by .0192 units on depression at follow-up due to the effect of harassment on perceived academic failure at followup, and the subsequent effect of perceived academic failure at follow-up on depression at follow-up.

What concerns might you have about making causal claims related to this specific mediation analysis?

While we did control for some variables, if there are other confounders (common causes) of any of the variables involved, this could bias our indirect effect and limit our ability to make causal claims. Additionally, because the mediator and outcome are measured at the same time, there's no way to know which comes first and this could make our indirect effect meaningless.

This is the end of the homework.